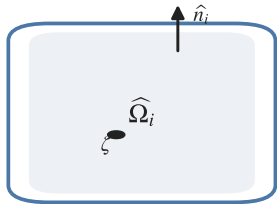
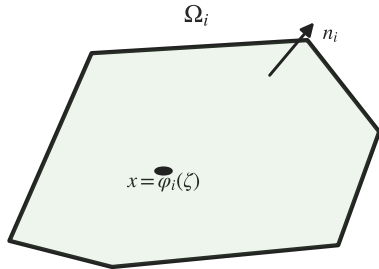


Reference patch



Physical patch


 $\varphi_i, \mathbf{J}_i$

Gradient pullback

$$\nabla_x u = \nabla_\zeta \hat{u}_i \mathbf{J}_i^{-1}$$

Piola flux

$$\hat{\mathcal{P}}_i = j_i \hat{\mathcal{G}}_i \mathbf{J}_i^{-T}$$

Normal map

$$n_i = \mathbf{J}_i^{-T} \hat{n}_i / \|\mathbf{J}_i^{-T} \hat{n}_i\|$$